

The Biophoton and Ultra-Weak Photon Emission Field

State of the field. A structured map for the OSF open-science book.

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• Overview

The map covers **18,355 works** and **39,312 authors**, grown by 2-hop citation expansion from **245 resolved Cifra seed works**. Bibliographic-coupling community detection (Leiden) separates the field into distinct intellectual sub-fields; the sonoluminescence/cavitation physics forms its **own** communities, confirming empirically that it is an adjacency to, not part of, the biophoton core.

Field-wide openness: **38%** of works are open access; **7301** have a preprint; **29** carry an open-data/code signal in their abstract.

• The sub-fields

UPE / biophoton core (community 0).

- **5,193 works**, 125 seeds, median year 2012, range 1922-2026.
- **Themes:** Biofield Effects and Biophysics, bioluminescence and chemiluminescence research, Photoreceptor and optogenetics research, Electromagnetic Fields and Biological Effects, Neural dynamics and brain function.
- **Key authors (seed-anchored):** Michael A. Persinger, Michal Cifra, Roeland Van Wijk, Jack A. Tuszyński, Francesco Musumeci, Agata Scordino.
- **Where it lives:** University of Catania (124); Laurentian University (107); Tufts University (104); Czech Academy of Sciences, Institute of Photonics and Electronics (98).
- **Geography (author affiliations):** US (982), CN (653), IT (535), CA (314), JP (245).
- **Mean author openness:** 0.46.
- **Representative works:**
 - Evidence for wavelike energy transfer through quantum coherence in photosynthetic systems (2007.0, 3335 cites)
 - Near-infrared fluorophores for biomedical imaging (2017.0, 2935 cites)
 - Lessons from nature about solar light harvesting (2011.0, 1924 cites)
 - Coherently wired light-harvesting in photosynthetic marine algae at ambient temperature (2010.0, 1663 cites)

Inside the biophoton core: sub-strands (Leiden res 3.0).

Re-clustering community 0 resolves the true UPE core into distinct research strands (the biofield/EEG/EMF/quantum-information literature splits off into its own sub-clusters):

- **human UPE & measurement methodology** (sub 1: 335 works, 44 seeds): Roeland Van Wijk, Michal Cifra, Eduard P.A. Van Wijk, Eduard van Wijk. e.g. Ultra-weak photon emission from biological samples: Definition, mechanisms, prop (2014.0, 269c)
- **cellular communication through light** (sub 9: 205 works, 14 seeds): Max Myakishev-Rempel, Ilya Volodyaev, Ivan Savelev, Daniel Fels. e.g. Eukaryotic Cells and their Cell Bodies: Cell Theory Revised (2004.0, 146c)
- **delayed luminescence & photosynthesis** (sub 4: 247 works, 12 seeds): Agata Scordino, Francesco Musumeci, Rosaria Grasso, M. Gulino. e.g. Monitoring and screening plant populations with combined thermal and chlorophyll (2006.0, 241c)
- **biophoton imaging & plant/animal UPE** (sub 2: 288 works, 7 seeds): Cristiano M. Gallep, Setsuko Komatsu, Kimihiko Kato, Hiroyuki Iyozumi. e.g. In vivo imaging of spontaneous ultraweak photon emission from a rat's brain corr (1999.0, 188c)
- **ROS / singlet-oxygen photon emission** (sub 0: 474 works, 7 seeds): Roeland Van Wijk, Eduard P.A. Van Wijk, Paul S. Francis, Neil W. Barnett. e.g. Singlet Molecular Oxygen Reactions with Nucleic Acids, Lipids, and Proteins (2019.0, 638c)
- **neural / brain optical signalling** (sub 15: 155 works, 6 seeds): Jiawei Dai, Christoph Simon, Zhuo Wang, Vahid Salari. e.g. Optical windows for head tissues in near-infrared and short-wave infrared region (2018.0, 204c)
- **coherence theory (Popp school)** (sub 19: 114 works, 5 seeds): Fritz-Albert Popp, F. A. Popp, L. V. Belousov, Jiin-Ju Chang. e.g. Energy Flow in Proteins (2008.0, 316c)
- **water & coherent domains** (sub 5: 241 works, 5 seeds): Pierre Madl, Larissa Brizhik, B.Л. Воейков, Alberto Tedeschi. e.g. Water Dynamics at the Root of Metamorphosis in Living Organisms (2010.0, 160c)
- **retinal & brain single-photon signalling** (sub 18: 116 works, 3 seeds): István Bókkon, Vahid Salari, Jack A. Tuszyński, Jiawei Dai. e.g. Single-photon detection by rod cells of the retina (1998.0, 271c)

Split-off, consciousness/paranormal wing (sub 8: 214 works, 9 seeds): Michael A. Persinger, Blake T. Dotta, Nirosha J. Murugan, Lukasz M. Karbowski. This is the reputationally-mixed strand the book should handle separately; researchers here are flagged `consciousness_adjacent=1` in `researchers.csv`.

ROS / redox / biochemiluminescence (community 3).

- **1,974 works**, 9 seeds, median year 2011, range 1969-2026.
- **Themes:** Mitochondrial Function and Pathology, Skin Protection and Aging, Redox biology and oxidative stress, Nitric Oxide and Endothelin Effects, Plant Stress Responses and Tolerance.
- **Key authors (seed-anchored):** Jin-Ming Lin, Hui Chen, Murray Grant, Franz Stäb, Zhen Lin, Krzysztof Sasak.
- **Where it lives:** Tsinghua University (7); Medical University of Lodz (7); Beiersdorf (Germany) (7); Royal Holloway University of London (6).
- **Geography (author affiliations):** CN (28), DE (20), GB (17), US (14), PL (12).
- **Mean author openness:** 0.47.

• **Representative works:**

- Peroxidase-dependent apoplastic oxidative burst in Arabidopsis required for pathogen resistance (2006.0, 556 cites)
- The Functional Organization and Control of Plant Respiration (2006.0, 504 cites)
- Arabidopsis systemic immunity uses conserved defense signaling pathways and is mediated by ja (2007.0, 449 cites)
- Plant mitochondria: past, present and future (2021.0, 361 cites)

sonochemistry / acoustic cavitation (community 1).

- **3,016 works**, 13 seeds, median year 2014, range 1971-2026.
- **Themes:** Ultrasound and Cavitation Phenomena, Advanced oxidation water treatment, Ultrasound and Hyperthermia Applications, Nanoplatfroms for cancer theranostics, Innovative Microfluidic and Catalytic Techniques Innovation.
- **Key authors (seed-anchored):** Oualid Hamdaoui, Slimane Merouani, Muthupandian Ashokkumar, Aissa Dehane, Kyuichi Yasui, Franz Grieser.
- **Where it lives:** Constantine 1 University (133); The University of Melbourne (109); King Saud University (100); National Institute of Advanced Industrial Science and Technology (87).
- **Geography (author affiliations):** CN (253), DZ (192), JP (190), US (181), GB (165).
- **Mean author openness:** 0.33.
- **Representative works:**
 - Sonochemical synthesis of amorphous iron (1991.0, 1249 cites)
 - The Temperature of Cavitation (1991.0, 1189 cites)
 - The characterization of acoustic cavitation bubbles: An overview (2010.0, 910 cites)
 - Remediation of soil and water contaminated with petroleum hydrocarbon: A review (2019.0, 878 cites)

bubble & fluid physics (community 2).

- **2,869 works**, 1 seeds, median year 2015, range 1936-2026.
- **Themes:** Ultrasound and Cavitation Phenomena, Ultrasound and Hyperthermia Applications, Cavitation Phenomena in Pumps, Minerals Flotation and Separation Techniques, Fluid Dynamics and Mixing.
- **Key authors (seed-anchored):** Detlef Lohse, Michel Versluis, Robert Mettin, Claus-Dieter Ohl, Nico de Jong, Ferenc Hegedűs.
- **Where it lives:** University of Twente (63); Max Planck Institute for Dynamics and Self-Organization (39); University of Göttingen (30); Budapest University of Technology and Economics (29).
- **Geography (author affiliations):** CN (289), US (180), NL (137), DE (134), FR (62).
- **Mean author openness:** 0.39.
- **Representative works:**
 - Microscale acoustofluidics: Microfluidics driven via acoustics and ultrasonics (2011.0, 937 cites)
 - Passive ultrasonic irrigation of the root canal: a review of the literature (2007.0, 842 cites)
 - Surface nanobubbles and nanodroplets (2015.0, 836 cites)

- Physics of bubble oscillations (2010.0, 805 cites)

sonoluminescence physics (community 4).

- **1,154 works**, 34 seeds, median year 2007, range 1951-2026.
- **Themes:** Ultrasound and Cavitation Phenomena, Quantum Electrodynamics and Casimir Effect, Nuclear Physics and Applications, Ultrasound and Hyperthermia Applications, Electrohydrodynamics and Fluid Dynamics.
- **Key authors (seed-anchored):** Kyuichi Yasui, Detlef Lohse, Seth Putterman, Kenneth S. Suslick, Weizhong Chen, An Yu².
- **Where it lives:** University of Illinois Urbana-Champaign (52); National Institute of Advanced Industrial Science and Technology (40); Institute of Catalysis and Petrochemistry (38); Max Planck Institute for Dynamics and Self-Organization (30).
- **Geography (author affiliations):** US (369), CN (306), RU (139), DE (136), JP (126).
- **Mean author openness:** 0.31.
- **Representative works:**
 - Electrogenenerated Chemiluminescence and Its Biorelated Applications (2008.0, 2113 cites)
 - Plasma, liquid interactions: a review and roadmap (2016.0, 1663 cites)
 - Non-thermal plasmas in and in contact with liquids (2009.0, 1266 cites)
 - Single-bubble sonoluminescence (2002.0, 1035 cites)

nanobubbles (community 6).

- **141 works**, 2 seeds, median year 2021, range 1975-2026.
- **Themes:** Minerals Flotation and Separation Techniques, Ultrasound and Cavitation Phenomena, nanoparticles nucleation surface interactions, Industrial Gas Emission Control, Nonlocal and gradient elasticity in micro/nano structures.
- **Key authors (seed-anchored):** Kyuichi Yasui, Lijuan Zhang, Jun Hu, Toru Tuziuti, Wataru Kanematsu, Limin Zhou.
- **Where it lives:** National Institute of Advanced Industrial Science and Technology (7); Northeast Normal University (6); National Institute of Technology (5); Chinese Academy of Sciences (5).
- **Geography (author affiliations):** CN (38), JP (13), AU (4), KR (4), US (2).
- **Mean author openness:** 0.37.
- **Representative works:**
 - Mysteries of bulk nanobubbles (ultrafine bubbles); stability and radical formation (2018.0, 216 cites)
 - Generation and stability of bulk nanobubbles: A review and perspective (2021.0, 189 cites)
 - The effect of ultrasound on bulk and surface nanobubbles: A review of the current status (2021.0, 110 cites)
 - High temperature and pressure inside a dissolving oxygen nanobubble (2019.0, 64 cites)

• Timeline of the field

Works per 5-year bin (whole universe):

- 1900 to 1904: 1
- 1905 to 1909: 1
- 1910 to 1914: 3
- 1915 to 1919: 2
- 1920 to 1924: 12
- 1925 to 1929: 44
- 1930 to 1934: 67
- 1935 to 1939: 38
- 1940 to 1944: 19
- 1945 to 1949: 28
- 1950 to 1954: 61
- 1955 to 1959: 65
- 1960 to 1964: 90
- 1965 to 1969: 147
- 1970 to 1974: 190
- 1975 to 1979: 256
- 1980 to 1984: 417
- 1985 to 1989: 521
- 1990 to 1994: 759
- 1995 to 1999: 1348
- 2000 to 2004: 1834
- 2005 to 2009: 2495
- 2010 to 2014: 3132
- 2015 to 2019: 3081
- 2020 to 2024: 3006
- 2025 to 2029: 726

• The open-vs-closed analysis

Within the biophoton core (community 0, researchers with ≥ 3 in-universe works), the openness overlay separates who builds the field in the open from who encloses it:

Most open:

- S. McHugh: openness 0.83 (OA 100%, preprint 100%)
- Reneta Toshkova: openness 0.75 (OA 100%, preprint 100%)
- Patrick Rebentrost: openness 0.75 (OA 100%, preprint 100%)
- Christoph Marquardt: openness 0.75 (OA 100%, preprint 100%)
- Donatas Zigmantas: openness 0.75 (OA 100%, preprint 100%)

- Carmelo M. Vicario: openness 0.75 (OA 100%, preprint 100%)
- Robert J. Usselman: openness 0.75 (OA 100%, preprint 100%)
- Stefano Pirandola: openness 0.75 (OA 100%, preprint 100%)

Least open:

- Bernard O Williams: openness 0.05 (OA 0%)
- Maxim V. Trushin: openness 0.05 (OA 0%)
- Herbert A. Pohl: openness 0.05 (OA 0%)
- E. A. Vinogradov: openness 0.05 (OA 0%)
- Bernard Valeur: openness 0.05 (OA 0%)
- Lawrence J. Marnett: openness 0.05 (OA 0%)
- V. S. Letokhov: openness 0.05 (OA 0%)
- D. K. Krause: openness 0.05 (OA 0%)

• Gaps & whitespace

- **Author disambiguation:** OpenAlex splits some core authors (Van Wijk, Popp merged here; others flagged in `author_splits_flagged.json`): a manual pass would sharpen centrality/ranking.
- **Full-text open-data signals:** open-data/code detection here is abstract-level; scanning OA full text would raise recall.
- **The consciousness-adjacent wing** (Persinger, Dotta, Tuszyński in community 0) is reputationally mixed and worth separating editorially in the book.

• Deliverables

- `fieldmap.sqlite` + `parquet/CSV`: the queryable database.
- `researchers.csv`: ranked outreach table (internal email column).
- `knowledge_map.html`: interactive researcher map.
- `*.graphml`: all four graphs for Gephi.